

Business Summary

What Matters

RVM008 is the main reliability concern. Its average downtime is about 50.6 minutes/day, more than twice the next-highest machine, with several days of particularly long downtime.

Fill level is the clearest indicator of collection need. Machines that need collection are generally over 80% full, compared with less than 50% for machines that don't.

Material rejection has two different problems:

- Unknown items: About 5% of transactions (1,756 items) couldn't be identified and were rejected 100% of the time.
- Aluminium: Even when identified, aluminium is rejected around 25% of the time, compared with only 6.3% for PET.

What We Recommend

Use the model to help prioritize collections, while improving the data behind it.

Collect more data before relying on the model at a larger scale

What We Should Watch Next

Before scaling to hundreds of machines, improve the data and keep monitoring the model.

- Truck and route information: Distance, collection time, truck capacity, and machine locations would allow us to move from predicting need to planning efficient routes.
- Sensor health monitoring: The system should be able to distinguish a genuinely full machine from a faulty sensor.
- Material rejection: Investigate why Aluminium and Unknown materials are rejected and whether this is causing user complaints.
- More history: Collect data across seasons, holidays, locations, and different machine types before assuming the model will behave the same way everywhere.